BSCS 3rd Semester

Probability and Statistics

Lecture Notes | Lectures 01 to 10

Contents covered

Basic definitions, events and their types, probability, statistics and data, conditional probability and tree method, card problems, multiplication law, random variables, counting rules (multiplication, permutation, combination), additive law, discrete and continuous random variables with mean and variance, and Bayes theorem.

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Lecture 01 — Introduction

Basic definitions

Random Experiment

Random experiment is an action or process that can be repeated under the same conditions, but the outcome cannot be predicted with certainty.

Exp

Tossing a coin: H, T.

Sample space

The total outcome from any experiment is called sample space (the set of all possible outcomes).

Exp

$S = \{H, T\}$

Event

A subset of sample space is called event.

Exp

Two coins are tossed: $S = \{HH, HT, TH, TT\}$
$A = \{HH\}$: only head appears.
$B = \{TT\}$: only tail appears.
$C = \{HT, TH\}$: both head and tail appear.

Types of event

There are two types of event.

i. Mutually exclusive event

ii. Not mutually exclusive event

Mutually exclusive event

Two events are mutually exclusive if they cannot occur at the same time.

Exp

When a coin is tossed, head or tail appears but not both.
For a die: $S = \{1,2,3,4,5,6\}, \quad A = \{1,3,5\}, \quad B = \{2,4,6\}$
$A \cap B = \phi$. These are M.E.

Not mutually exclusive event

Two events are not mutually exclusive if they can occur at the same time.

Exp

$S = \{1, 2, 3, 4, 5, 6\}, \quad A = \{1, 3, 5, 6\}, \quad B = \{2, 4, 6\}$
$A \cap B = \{6\}$

Lecture 02 & 03 — Probability, Statistics and Data

Probability

Probability is the measure of chance or likelihood that a particular event occurs.

$$P(A) = \frac{n(A)}{n(S)}$$

Here $n(A)$ is the number of outcomes in the **event** and $n(S)$ is the number of outcomes in the **sample space**.

Exp

$S = \{1, 2, 3, 4, 5, 6\}$, so $n(S) = 6$
$A = \{2, 4, 6\}$, so $n(A) = 3$
So, $\quad P(A) = \frac{3}{6} = \frac{1}{2} = 0.5$

MCQ

Probability lie between 0 and 1.

$$0 \leq P(A) \leq 1$$

Statistics

It is the branch of mathematics that deals with data: collecting, analyzing, interpreting and presenting it as useful information, or making effective decisions.

Exp

Use in education to analyze student performance.
Use in business.

Data

The collection of raw facts, numbers and observations.

Exp

Marks of students...

Data Analysis

Data analysis means studying the collected data.

Exp

If you take the marks of students and find the average, you are doing data analysis.

Types of statistics

i. Descriptive statistics

ii. Inferential statistics

Descriptive statistics

It is used to summarize and describe the main features of data.

Exp

Mean, median, mode

Inferential statistics

It is used to make a conclusion about a large group based on data from a sample.

Exp

If a man starts a business of chocolate, he chooses some people from a large population to taste his chocolate and makes a conclusion.

Types of data

i. Discrete data

ii. Continuous data

Discrete data

Countable data is called discrete data. It includes only countable values, such as whole numbers $\{0, 1, 2, 3, \ldots\}$.

Exp

Number of students in a class.

Continuous data

Measurable data is called continuous data. It may be in decimal form.

Exp

Height of a student (4.5 ft)
Weight of a student (60 kg)

Population

Population means the entire group of individuals or items that you want to study.

Exp

All students of a school.
All people living in Pakistan.

Q: Probability of two dice

$$S = \left\{ \begin{array}{cccccc} (1,1) & (1,2) & (1,3) & (1,4) & (1,5) & (1,6) \\ (2,1) & (2,2) & (2,3) & (2,4) & (2,5) & (2,6) \\ (3,1) & (3,2) & (3,3) & (3,4) & (3,5) & (3,6) \\ (4,1) & (4,2) & (4,3) & (4,4) & (4,5) & (4,6) \\ (5,1) & (5,2) & (5,3) & (5,4) & (5,5) & (5,6) \\ (6,1) & (6,2) & (6,3) & (6,4) & (6,5) & (6,6) \end{array} \right\}$$

Event A: both dice show the same number.

$$A = \{(1,1),(2,2),(3,3),(4,4),(5,5),(6,6)\}$$

$$n(A) = 6, \qquad n(S) = 36$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{6}{36} = \frac{1}{6} \approx 0.167$$

Probability of two coins

$S = \{HH, HT, TH, TT\}$

Event A: first coin shows head. $A = \{HH, HT\}$

$$n(A) = 2, \qquad n(S) = 4$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{2}{4} = \frac{1}{2} = 0.5$$

Lecture 04 — Conditional Probability

Conditional Probability

Conditional probability is the probability of an event occurring when another event has already occurred.
It is written as: $P(A \mid B)$ = probability of A given that B.

Formula

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

$P(A \cap B)$ = probability when both A and B occurred.

Exp

When 3 fair coins are tossed, then find the probability.
- At least two tails appear
- first coin shows head.

$S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}, \quad n(S) = 8$

Event A: at least two tails appear.

$$A = \{HTT, THT, TTH, TTT\}, \qquad n(A) = 4$$

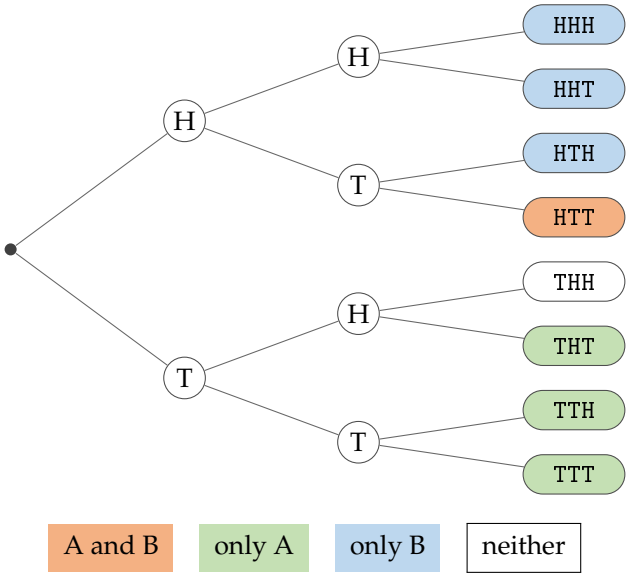
$$P(A) = \frac{n(A)}{n(S)} = \frac{4}{8} = \frac{1}{2}$$

Event B: first coin shows head.

$$B = \{HHH, HHT, HTH, HTT\}, \qquad n(B) = 4$$

$$P(B) = \frac{n(B)}{n(S)} = \frac{4}{8} = \frac{1}{2}$$

Tree method



Q

At least two tails appear but condition is first coin shows head.

$$A \cap B = \{HTT\}, \qquad n(A \cap B) = 1$$

$$P(A \cap B) = \frac{n(A \cap B)}{n(S)} = \frac{1}{8} = 0.125$$

For conditional probability:

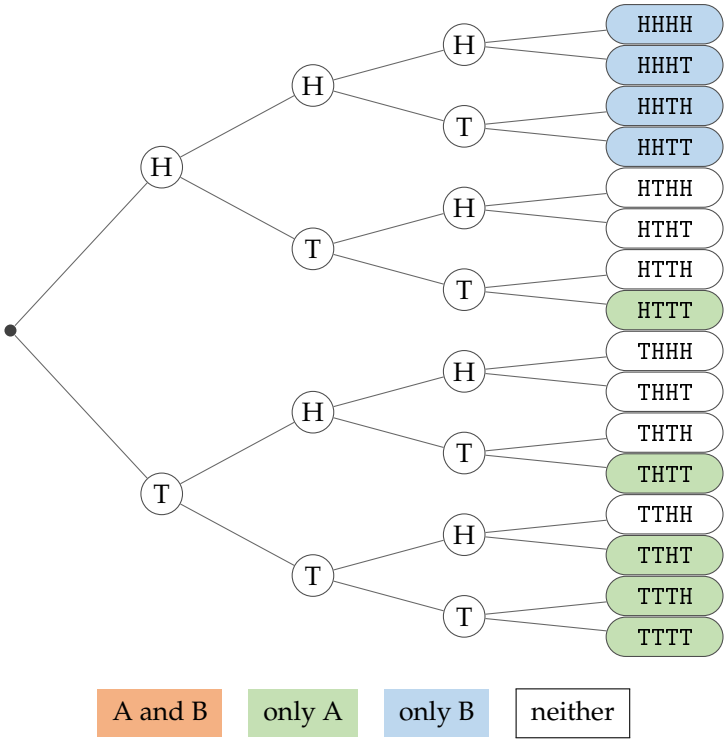
$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{1/8}{1/2}$$

$$\boxed{P(A \mid B) = \frac{1}{4} = 0.25}$$

Q

When 4 fair coins are tossed then find the probability.
- At least three tails appear
- first two coins show head.

Tree method



$$S = \{HHHH, HHHT, HHTH, HHTT, HTHH, HTHT, HTTH, HTTT,$$
$$THHH, THHT, THTH, THTT, TTHH, TTHT, TTTH, TTTT\}$$

$$n(S) = 16$$

Event A: at least three tails appear.

$$A = \{HTTT, THTT, TTHT, TTTH, TTTT\}, \qquad n(A) = 5$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{5}{16} \approx 0.31$$

Event B: first two coins show head.

$$B = \{HHHH, HHHT, HHTH, HHTT\}, \qquad n(B) = 4$$

$$P(B) = \frac{n(B)}{n(S)} = \frac{4}{16} = \frac{1}{4} = 0.25$$

$$A \cap B = \phi, \qquad n(A \cap B) = 0$$

$$P(A \cap B) = \frac{n(A \cap B)}{n(S)} = \frac{0}{16} = 0$$

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{0}{1/4}$$

$$\boxed{P(A \mid B) = 0}$$

Lecture 05 & 06 — Examples of Probability and Multiplication Law

Example of Probability

A card is drawn from an ordinary deck of 52 playing cards. Find probability:

1. card is Red
2. card is Diamond
3. card is 10.

MCQ's

- Red = 13 Diamond and 13 Heart. Total Red = 26
- Black = 13 Spade and 13 Club. Total Black = 26
- Face card = 12 (K, Q, J of each of the 4 suits) Total card = 52

Each suit has 13 cards: A, 2, 3, 4, 5, 6, 7, 8, 9, 10, J, Q, K.

Solution

$n = 52, \quad n(S) = 52$

1) Let A be the event in which card is Red.

$$n(A) = 26, \qquad P(A) = \frac{n(A)}{n(S)} = \frac{26}{52} = \frac{1}{2}$$

2) Let B be the event in which card is Diamond.

$$n(B) = 13, \qquad P(B) = \frac{n(B)}{n(S)} = \frac{13}{52} = \frac{1}{4} = 0.25$$

3) Let C be the event in which card is 10.

$$n(C) = 4, \qquad P(C) = \frac{n(C)}{n(S)} = \frac{4}{52} = \frac{1}{13} \approx 0.077$$

Imp Question: Multiplication law / Rule / Product law

Statement

If A and B are the two events defined in sample space S then

- $P(A \cap B) = P(A) \cdot P(B \mid A)$
- $P(A \cap B) = P(B) \cdot P(A \mid B)$

Proof

By the def of conditional probability:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} \qquad \therefore\ P(B) \neq 0$$

$$P(A \cap B) = P(B) \cdot P(A \mid B)$$

Now,

$$P(B \mid A) = \frac{P(A \cap B)}{P(A)} \qquad \therefore\ P(A) \neq 0$$

$$P(A \cap B) = P(A) \cdot P(B \mid A)$$

Lecture 07 — Random Variable and Counting Rules

Random variable / Chance variable

Random variable

A variable whose values are determined by the outcome of a random experiment.

Exp

Two coins are tossed: $S = \{HH, HT, TH, TT\}$.
When a numerical value is assigned to each outcome (number of heads), $X = 0, 1, 2$.

$$HH \to 2, \qquad HT \to 1, \qquad TH \to 1, \qquad TT \to 0$$

It is also called Stochastic variable.

Counting sample point

Counting sample points are divided into three rules. These rules are as follows:

1. Rule of Multiplication
2. Rule of Permutation
3. Rule of Combination

Rule of Multiplication

If two experiments with m and n outcomes occur, then together they have

$$m \times n = mn \ \to \text{ outcomes}$$

Example

Coin: $m = \{H, T\}$, so $m = 2$
Dice: $n = \{1, 2, 3, 4, 5, 6\}$, so $n = 6$
Then by multiplication rule: $m \times n = 2 \times 6 = 12$ outcomes

Rule of Permutation

In which order of selection does matter.

$$^{n}P_r = \frac{n!}{(n-r)!}$$

$n \to$ Total objects (observation), $\qquad r \to$ Selection

Example

Members in club $n = 4$; President, Secretary, Treasurer $r = 3$

$$^4P_3 = \frac{4!}{(4-3)!} = \frac{4!}{1!} = \frac{4 \cdot 3 \cdot 2 \cdot 1}{1} = 24 \quad \text{Ans.}$$

Rule of Combination

In which order of selection does not matter.

$$^nC_r = \binom{n}{r} = \frac{n!}{r!\,(n-r)!}$$

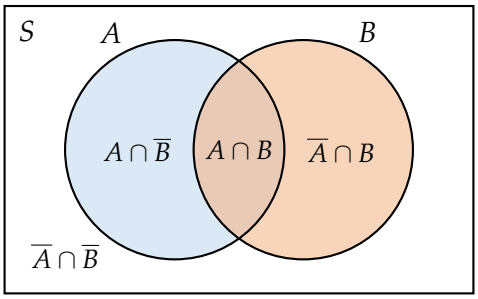
Example

List committee $n = 4$; choose $r = 3$

$$\binom{4}{3} = \frac{4!}{3!\,(4-3)!} = \frac{4!}{3!\,1!} = \frac{24}{6} = 4$$

Additive Law

If A and B are two not mutually exclusive events defined in S



$$\boxed{P(A \cup B) = P(A) + P(B) - P(A \cap B)}$$

Note. This law holds for any two events. If A and B are mutually exclusive, $P(A \cap B) = 0$ and it reduces to $P(A \cup B) = P(A) + P(B)$.

Proof

The event $A \cup B$ is written as:

$$A \cup B = A \cup (\overline{A} \cap B)$$

$$P(A \cup B) = P(A) + P(\overline{A} \cap B) \tag{1}$$

Again event B is decomposed:

$$B = (A \cap B) \cup (B \cap \overline{A})$$

$$P(B) = P(A \cap B) + P(B \cap \overline{A}) \tag{2}$$

Subtract (2) from (1):

$$P(A \cup B) - P(B) = P(A) + P(\overline{A} \cap B) - P(A \cap B) - P(\overline{A} \cap B)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Lecture 09 — Discrete and Continuous Random Variables

Discrete Random variable

Whole number (countable values).

$$\text{Mean} = E(x) = \sum x\,p$$

$$\text{Variance} = E(x^2) - \left[E(x)\right]^2, \qquad \therefore\ E(x^2) = \sum x^2\,p$$

$$S.D = \sqrt{Var(x)}$$

Example 1

A discrete random variable has probability given below:

$X =$	-2	-1	0	1	2	3
$P(\ ) =$	0.2	k	0.1	$2k$	0.1	$2k$

Solution

$$0.2 + k + 0.1 + 2k + 0.1 + 2k = 1 \qquad \therefore \sum p = 1$$

$$0.4 + 5k = 1 \quad \Rightarrow \quad 5k = 1 - 0.4 \quad \Rightarrow \quad k = \frac{0.6}{5}$$

$$k = 0.12$$

$$\sum xp = (-2)(0.2) + (-1)(0.12) + (0)(0.1) + (1)(2)(0.12) + 2(0.1) + 3(2)(0.12)$$

$$\boxed{\text{Mean} = 0.64}$$

$$\text{Variance} = E(x^2) - \left[E(x)\right]^2 = 3.72 - (0.64)^2$$

$$\boxed{\text{Variance} = 3.3104}$$

$$S.D = \sqrt{3.3104} \qquad \boxed{S.D = 1.819}$$

$E(x^2) = \sum x^2 p = 4(0.2) + 1(0.12) + 0 + 1(0.24) + 4(0.1) + 9(0.24) = 3.72$. This value is used in the notes without the working.

Example 2

$X =$	0	1	2
$P =$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$

$$\text{Mean} = E(x) = \sum xp = (0)\left(\tfrac{1}{4}\right) + (1)\left(\tfrac{1}{2}\right) + (2)\left(\tfrac{1}{4}\right)$$

$$\boxed{E(x) = 1}$$

$$\text{Variance} = E(x^2) - \left[E(x)\right]^2, \qquad E(x^2) = \sum x^2 p = 0 + \tfrac{1}{2} + 1 = 1.5$$

$$= 1.5 - 1 = 0.5 \qquad \boxed{Var = \tfrac{1}{2}}$$

$$S.D = \sqrt{Var(x)} = \sqrt{\tfrac{1}{2}} \qquad \boxed{S.D = \frac{1}{\sqrt{2}} \approx 0.707}$$

Continuous Random Variable

$$\text{Mean} = E(x) = \int x\, f(x)\, dx$$

$$\text{Variance} = E(x^2) - \left[E(x)\right]^2, \qquad \therefore\ E(x^2) = \int x^2 f(x)\, dx$$

Example

$f(x) = k(x - x^2), \qquad 0 \le x \le 1$
To find: Mean = ? k = ? Variance = ?

Solution

For $f(x)$ to be a probability density function, the total area must be 1.

$$k \int_0^1 (x - x^2)\, dx = 1$$

$$k \left[\int_0^1 x\, dx - \int_0^1 x^2\, dx \right] = 1$$

$$k \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^1 = 1$$

$$k \left[\frac{(1)^2}{2} - \frac{(1)^3}{3} \right] = 1$$

$$k \left(\frac{1}{2} - \frac{1}{3} \right) = 1 \ \Rightarrow\ k \left(\frac{3-2}{6} \right) = 1$$

$$k \left(\frac{1}{6} \right) = 1 \ \Rightarrow\ \boxed{k = 6}$$

Mean:

$$E(x) = \int_0^1 x\, f(x)\, dx = \int_0^1 x\big(6(x - x^2)\big) dx = 6 \int_0^1 (x^2 - x^3)\, dx$$

$$= 6 \left[\frac{x^3}{3} - \frac{x^4}{4} \right]_0^1 = 6 \left(\frac{1}{3} - \frac{1}{4} \right) = \frac{1}{2}$$

$$E(x^2) = \int_0^1 x^2 \big(6(x - x^2)\big) dx = 6 \left[\frac{x^4}{4} - \frac{x^5}{5} \right]_0^1 = 6 \left(\frac{1}{4} - \frac{1}{5} \right) = \frac{3}{10}$$

The two integrals above (mean and $E(x^2)$) are not written out in the handwritten notes; the next page starts directly with their values $\frac{1}{2}$ and $\frac{3}{10}$. The working is filled in here.

Variance:

$$\text{Variance} = E(x^2) - \left[E(x)\right]^2 = \frac{3}{10} - \left(\frac{1}{2} \right)^2 = \frac{3}{10} - \frac{1}{4}$$

$$\boxed{Var = \frac{1}{20}}$$

Lecture 10 — Bayes Theorem

Bayes theorem

If the events $A_1, A_2, A_3, \ldots, A_k$ form a partition of a sample space S, that is the events A_i are mutually exclusive and their union is S, such that an event B can occur only if one of the A_i occurs, then for any i

$$P(A_i \mid B) = \frac{P(A_i)\,P(B \mid A_i)}{\sum_{i=1}^{k} P(A_i)\,P(B \mid A_i)}, \qquad \text{for } i = 1, 2, \ldots, k$$

Proof

By the definition of conditional probability, the events are A_i and B:

$$P(A_i \cap B) = P(A_i) \cdot P(B \mid A_i) \tag{i}$$
$$P(A_i \cap B) = P(B) \cdot P(A_i \mid B) \tag{ii}$$

By comparing eq (i) and (ii):

$$P(A_i) \cdot P(B \mid A_i) = P(B) \cdot P(A_i \mid B)$$

$$P(A_i \mid B) = \frac{P(A_i) \cdot P(B \mid A_i)}{P(B)} \tag{A}$$

We can write event B.

$$\begin{aligned} B &= S \cap B \\ &= (A_1 \cup A_2 \cup A_3 \cup \ldots \cup A_k) \cap B \\ B &= (A_1 \cap B) \cup (A_2 \cap B) \cup \ldots \cup (A_k \cap B) \end{aligned}$$

$$\begin{aligned} P(B) &= P\big[(A_1 \cap B) \cup (A_2 \cap B) \cup \ldots \cup (A_k \cap B)\big] \\ &= P(A_1 \cap B) + P(A_2 \cap B) + \ldots + P(A_k \cap B) \end{aligned}$$

$\therefore$ $A_i \cap B$ are mutually exclusive.

$$P(B) = P(A_1) \cdot P(B \mid A_1) + \ldots + P(A_k) \cdot P(B \mid A_k)$$

$$P(B) = \sum_{i=1}^{k} P(A_i) \cdot P(B \mid A_i)$$

By putting value of $P(B)$ in eq (A), we get

$$\boxed{P(A_i \mid B) = \frac{P(A_i) \cdot P(B \mid A_i)}{\sum_{i=1}^{k} P(A_i) \cdot P(B \mid A_i)}}$$